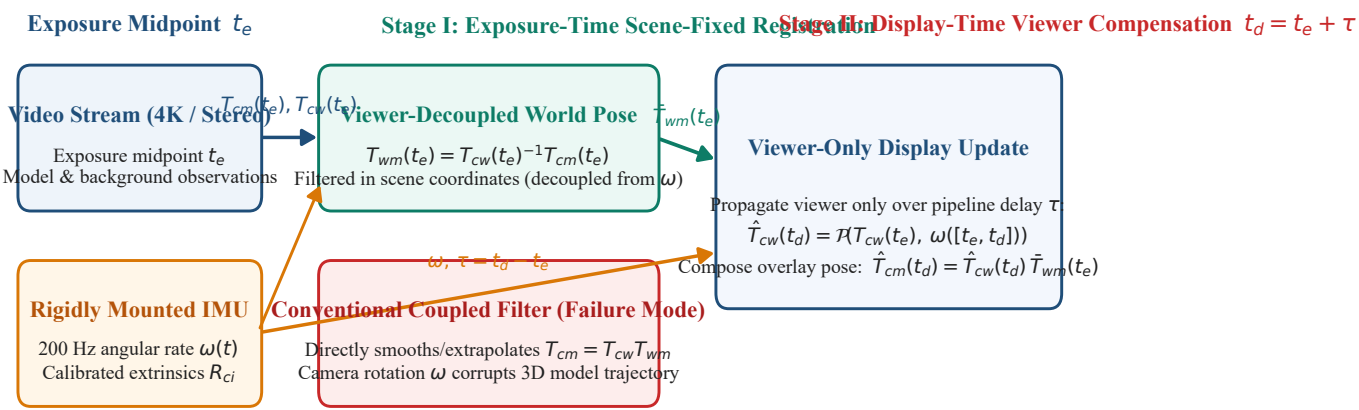

Exposure Midpoint t_e
Stage I: Exposure-Time Scene-Fixed Registration
Stage II: Display-Time Viewer Compensation $t_d = t_e + \tau$
Video Stream (4K / Stereo)
Exposure midpoint t_e
Model & background observations
$T_{cm}(t_e),\ T_{cw}(t_e)$
Viewer-Decoupled World Pose
$T_{wm}(t_e) = T_{cw}(t_e)^{-1} T_{cm}(t_e)$
Filtered in scene coordinates (decoupled from ω)
$\bar{T}_{wm}(t_e)$
Viewer-Only Display Update
Propagate viewer only over pipeline delay τ:
$\hat{T}_{cw}(t_d) = \mathcal{P}(T_{cw}(t_e),\ \omega([t_e, t_d]))$
Compose overlay pose: $\hat{T}_{cm}(t_d) = \hat{T}_{cw}(t_d)\, \bar{T}_{wm}(t_e)$
Rigidly Mounted IMU
200 Hz angular rate $\omega(t)$
Calibrated extrinsics R_{ci}
$\omega,\ \tau = t_d - t_e$
Conventional Coupled Filter (Failure Mode)
Directly smooths/extrapolates $T_{cm} = T_{cw} T_{wm}$
Camera rotation ω corrupts 3D model trajectory